Introduction

Need Statement: I want to reduce my household food waste.

Design Goals

- **Visibility:** users should be able to discern and access relevant information and features quickly and easily
- **Efficiency:** users should be able to interact with frequently used features with minimal intermediate steps
- **Integration:** users should be able to integrate the design into their daily routine

Chosen Design Approach

Our final design is a mobile app that combines a grocery list, an inventory tracker, and a recipe finder. Since food is wasted on a daily basis, we needed something close to the user. Ultimately, we decided to use a mobile format because most people have their mobile devices on them almost at all times, which we believe lends support for users to more easily adopt the app into their daily routines. After we conducted the initial field work study, we identified two common reasons household food items are wasted:

1. Forgetting a food item exists in their pantry
2. Not knowing how to use a food item before it expires

To tackle these issues, we made the main page of our app a grocery list, a tool most people are familiar with, to create positive transfer when integrating it into their daily routines. Our app also includes a pantry page to remind users of the status of the expiration date and quantity of the food items they have. We added a recipe finder feature to address the concern about finding ways to use a food item before it expires. Using our app to find recipes will simplify the process of finding a recipe using ingredients you have more efficiently. To refine our design approach, we aggregated together related pages and removed or hid miscellaneous elements so as not to distract from our design's main objectives.

Conceptual Model Iterations

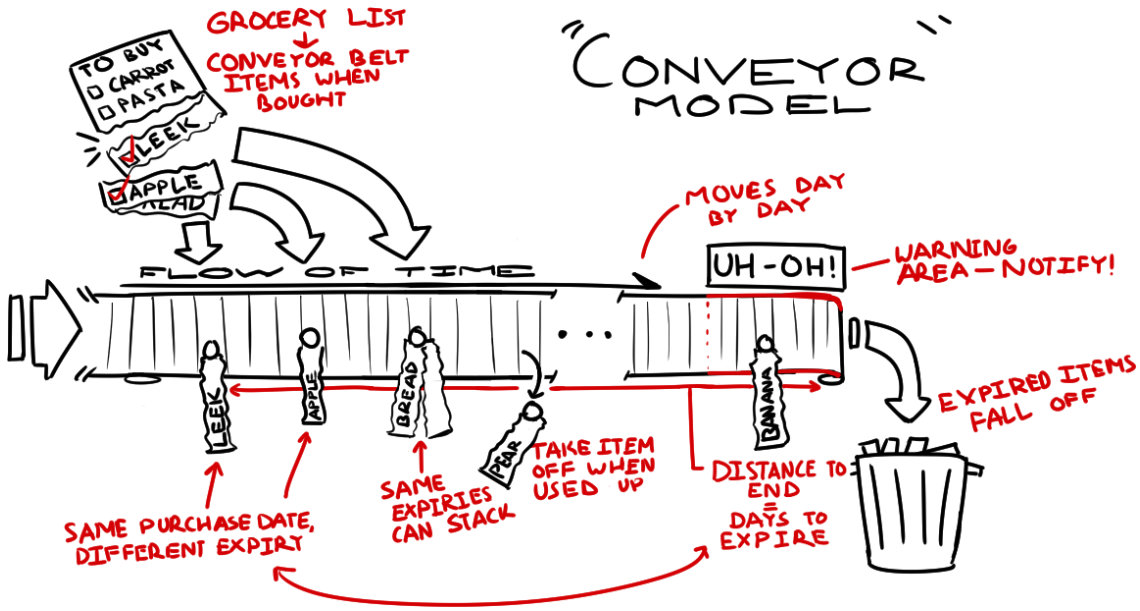


Figure 1: initial "Conveyor" conceptual model

The initial "Conveyor" conceptual model described candidate designs as a conveyor belt upon which grocery items would be attached, with their positions along the belt corresponding to their expiry date. The entire belt would be visible at all times and would steadily move placed items towards their expiry. Using the grocery items would remove them from the conveyor belt, as a fully consumed item can no longer expire.

While this conceptual model was good for explaining at the time-linked nature of the design's interactions, it was found during our cognitive walkthroughs (on a lo-fi prototype built on this model) to be lacking in its ability to connect with future intended features of the design, especially anything pertaining to our second user need relating to grocery item usage. While this was the only critique made towards our conceptual model (most of the issues stemmed from ineffective or missing signifiers), this issue led to the creation of the "Factory" conceptual model, which coupled the original "conveyor belt" metaphor with the addition of dedicated "stations" that represented our design's intended user tasks. This change, linked with the early low-fidelity proposal of each task having its own screen, recontextualized the conveyor belt from a representation of the flow of time to a representation of how the products of each task flowed into subsequent "stations". As this new conceptual model was built with the prototypes of the time in mind, there was no need to create a new prototype to embody the new conceptual model.

In our final design approach, the Groceries tab acts as a "bulletin board" in the factory, informing the other stations of what items should be purchased and thus brought into the factory. The Pantry tab acts as the "loading zone", where items requested by the "bulletin board" are loaded into the factory's conveyor belt and tracked for their progress. The Find Recipes page then

operates as the main factory floor, where items placed on the conveyor belt by the Pantry "loading zone" are used and their records updated, with expired items being funneled out of the factory as refuse.

Conceptual Model Summary: Following cognitive walkthrough feedback, the initial "Conveyor" model was expanded into the "Factory" model by adding task-specific stations, improving clarity for recipe-related tasks and aligning the conceptual flow with user goals.

Evaluation and Prototyping Methods

Research Question 1: Efficiency

- How efficient is our design when logging food items, reminding users about upcoming expiration dates, and finding recipe ideas?

It is important to assess how efficiently users can use our design to perform core tasks. If we can identify areas of hesitation, confusion, and redundancy, then we can identify which elements of our design to focus on and further iterate to improve task performance and user experience. In addition, increasing the efficiency of our design affords a more seamless experience for users, improving long term user retention, as they are less likely to give up after facing frustration.

Research Question 2: Integration

- How well could our design integrate into the user's usual food management routine?

It is important to assess how well-integrated our design could be in the user's daily routine because it allows us to understand when, how, and why users use our system. Additionally, insight and understanding of these questions will allow us to better design our system, such that it can more easily fit into the user's existing workflow. Furthermore, these questions serve as a guide to improving the product's ease of use and user experience in various use contexts, ultimately increasing the design's persuasiveness to new (or existing) users.

Research Question 3: Visibility

- How visible are the relevant information and intended actions that a user can perform when logging food items, checking upcoming expiration dates, or finding recipe ideas?

It is important to assess how visible our design is for users, especially regarding elements central to the core tasks of our design and its corresponding objectives. The data we gain from this research question allows us to learn what information the user visually perceives when completing these tasks and what information is missing or otherwise misinterpreted. This gives us the opportunity to iterate our design to better support users when completing core tasks, reduce erroneous behavior, and increase user retention.

Summary of Evaluation

Our usability tests were conducted as in-person studies, using a think-aloud observation followed by a questionnaire, with 11 participants above the age of 19. The session was recorded on Zoom if the participant consented to being recorded. If participants didn't consent to being recorded, moderators relied solely on the notes taken during observation. During the observation, participants were given a scenario and a set of tasks to perform using a think-aloud method. The study moderator observed their actions using a coding sheet. After the observation was completed, participants completed a questionnaire consisting of 5-point Likert scale and open-ended questions to further address our research questions.

Following feedback from the evaluation sign-off, we changed our evaluation method from an interview to a questionnaire, as we decided that a questionnaire would be better suited to collecting Likert scale data and addressing our evaluation goals. Initially, we had one research question that encompassed both efficiency and integration together, but we ended up splitting them into 2 distinct research questions as we felt that it would make our usability study more focused. We initially had two surveys but after piloting, we combined both surveys into one to streamline and simplify the study process. Minor changes were made to the prototype design, such as modifying visual elements and adjusting interaction flow to make it more intuitive, easier to use, and more interactive.

Evaluation Rationale

To minimize the daily burden on our users when using our application, we wanted to evaluate efficiency, defined as having minimum clicks on a given task. We measured efficiency through both objective click counts and subjective feedback to triangulate data and increase reliability of our findings. We evaluated integration because we didn't want our design to be a major disruption to a user's existing daily routine, as we believe that if our app does not easily integrate itself into a potential user's workflow, adoption would be less likely. Lastly, we wanted to evaluate visibility: our design should be as clear as possible while providing adequate feedback, information, and visual guidance for each task, reducing erroneous confusion and the cognitive load required to use our system.

Usability testing was conducted by asking participants to perform a set of tasks and observe their behaviour and actions while performing the task. This allowed us to observe how users interact with our design, such as number of clicks taken to complete a task, which parts of the screen were interacted with during a task, and any verbal comments or behavioural cues. The goal of our usability test was to identify usability problems in our prototype, identify opportunities to improve, and learn more about user's thought process and preference when interacting with our design. This provides us with valuable data which we can leverage to answer our research questions. Participants also completed a questionnaire where they provided subjective feedback

on their experience regarding the difficulty and efficiency of our design, the visibility of information, and how well-integrated the product could be in their daily lives. We decided to use a questionnaire to leverage Likert-scale questions which are best-suited at addressing user degrees of opinion on satisfaction and task performance. The choice to use a questionnaire also afforded us the option of adding open-ended questions to provide comprehensive feedback for each task that participants performed. The combination of a questionnaire and observation provides us multiple angles of exploration and allows us to triangulate our data and sufficiently address our research questions.

Prototyping Approach and Rationale

Prototyping objectives:

- Evaluating efficiency, integrability, and visibility of grocery item tracking and food item use finding
- Evaluating impact of design on recall of grocery item expiry dates

Initial prototyping followed the low-fidelity prototype created for the Shark Tank pitch which followed the 'Conveyor Belt' conceptual model (Fig. 1). The Figma prototype had five screens corresponding to distinct expected user tasks: grocery list management, transferring grocery list items to the pantry, searching for recipes using existing pantry items, expiry date management, and user settings. These tasks then informed our final 'Factory' conceptual model. However, internal evaluations found the design to be too cluttered and that some of the screens were too similar, making completing tasks confusing and repetitive. This prompted a new approach (and thus a new low-fidelity prototype) with reduced horizontality, using only two tabs representing the grocery list and pantry. We decided to focus on need-related features by integrating 'searching for recipes' and 'expiry date management' features into the two screens. After another internal evaluation, it was deemed sufficient to pilot and prototype.

Ultimately, we focused on a horizontal scope, as breadth across multiple features would lead to depth in overall usage, due to how our features are interconnected. Grocery list and pantry page traversal were prioritized, while interactions that used familiar elements (i.e. keyboards, counters) were simplified and fast-forwarded into a completed state. Grocery items that were not relevant to the evaluation task used placeholder text. To simplify usability testing, the design was divided into separate tasks, each with their own intermission screen for participants to wait for further instruction.

An improvement that could be made in the future is the amount of low-fidelity iteration before progression to medium-fidelity prototyping, as the initial low-fidelity prototype progressed into medium-fidelity before being completely scrapped, wasting time and work. In the future, more time should be spent ensuring that all members are on the same page of the lo-fi prototype

through communication, before proceeding to the mid-fi prototype. Spending more time creating a component library would have also helped alleviate later issues fixing inconsistencies between duplicated elements in separate screens.

Analysis – Findings:

Research Question 1: Efficiency - How efficient is our design when logging food items, reminding users about upcoming expiration dates, and finding recipe ideas?

Claim: Our design is reasonably efficient when logging food items and reminding users about expiration dates but could use some improvements especially when it comes to finding recipe ideas, which was shown to by far be the least efficient part of our design.

Evidence: Adjusting our analysis to account for a buffer of 3 clicks, where > 3 clicks from the ideal would indicate inefficiency, we can see that in the objective data compiled from the coding sheets filled out by examiners, logging food items is efficient while searching for recipes is inefficient. We can see the data supporting this. When we look at the average number of clicks it took for a participant to search for a recipe, we see that at ~8.9 clicks for completion, we are outside of our 3 click buffer from the ideal of 3 (Fig 4). Contrary to this, we see that with logging food items both tasks central to this objective fall within our click buffer (Fig 2 & 3). Our claim of logging food items being efficient is further supported by the questionnaire responses which have an >80% non-negative rating regarding efficiency. We employed a different type of analysis to determine that reminding users of expiration dates is efficient. The evidence that supports this is drawn from the fact that all 11 of our participants were able to notice the reminder notification (as noted in the coding sheets) and 8/11 of them indicated that our design would help remind them (stated themselves in the questionnaire responses).

Confidence: Our confidence varies on this research question. We note a high level of confidence when it comes to the logging of food items & reminders given to users on upcoming expiration dates because data compiled from both the coding sheets and questionnaire support a similar conclusion. On the other hand, we note a low level of confidence in terms of the efficiency for finding recipe ideas as although users claim they feel the design to be efficient from their responses to the questionnaire, our objective data analysis contradicts this.

Research Question 2: Integration - How well could our design be integrated into the user's usual food management routine?

Claim: Our design could integrate well for users who already track or itemize tasks but may introduce additional friction for general users who do not.

Evidence: Questionnaire data shows that although the majority of our participants (6/11) agree they would frequently use our design, there are still the 3 that are neutral/undecided (Fig 9). Thematic analysis of this data (Fig 10) alongside linking the Likert and explanation questionnaire responses together reveal that the main point of contention for the undecided users is that the design may be "too much work" suggesting friction introduced by our design into their daily routines. Only 2 participants disagreed that they would frequently use our design, of these both claimed that food waste was not a concern of theirs.

Confidence: High. Responses are consistent (among respective agreement group) across all participants

Research Question 3: Visibility - How visible are the relevant information and intended actions that a user can perform when logging food items, checking upcoming expiration dates, or finding recipe ideas?

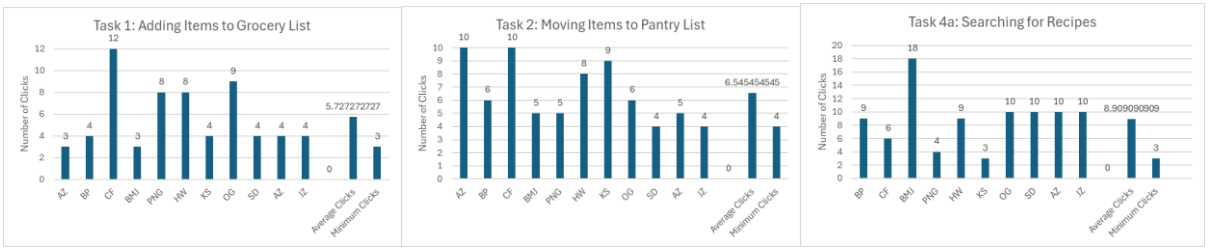
Claim: Our design exhibits good visibility for logging items and checking expiration dates but is less intuitive for recipe search and the marking of items as complete.

Evidence: Our questionnaire data overwhelmingly indicates that visibility is good regarding grocery and pantry features, with only 2/44 of related responses being negative for visibility (Fig 11). However, thematic analysis shows that confusion on visual elements core to searching for a recipe was a reoccurring concern, with several reports on aspects such as recipe search filters or the find buttons (Fig 6,8,12).

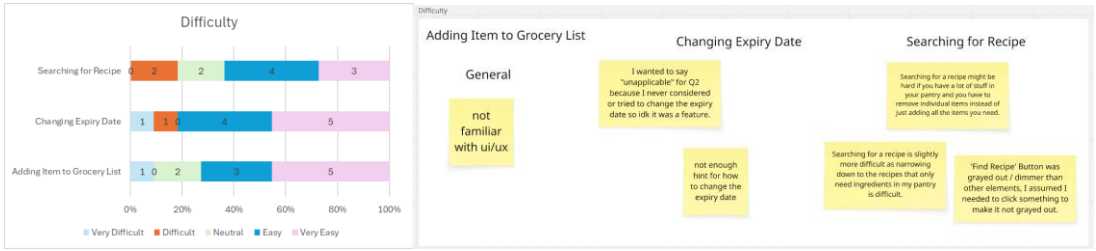
Additionally, observations made by examiners revealed concerns on the "check" completion buttons

Confidence: High. Observation data and questionnaire responses consistently identified visibility strengths and weaknesses across tasks. Furthermore, affinity diagrams and clustering also add additional context to what was observed.

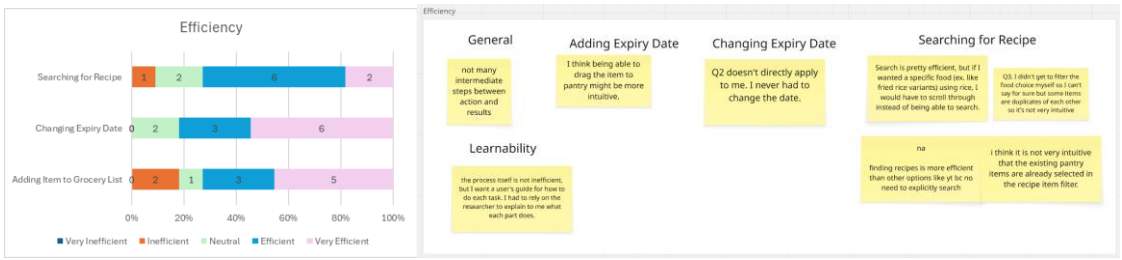
Data Visualizations



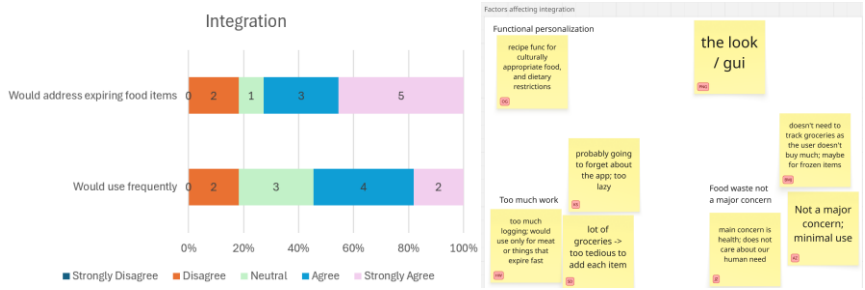
Figures 2, 3, 4: Number of clicks it took to complete each task (Objective Efficiency)



Figures 5, 6: Subjective Difficulty Analysis



Figures 7, 8: Subjective Efficiency Analysis



Figures 9, 10: Subjective Integration Analysis

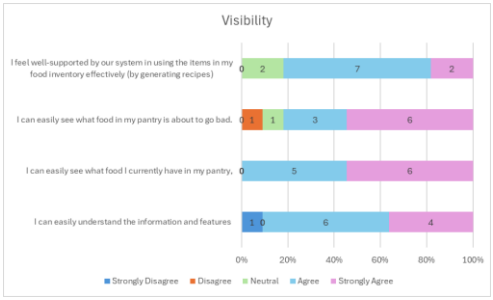


Figure 11: Subjective Visibility Analysis

Figure 12: Confusion with Design

Design Reflection and Recommendations

Although our prototype design wasn't fully functional, it was robust and usable. Most users were able to clearly navigate between all the screens with ease and understand how to perform the core functions of our design with minimal assistance and confusion. This was supported with the results of our usability testing; most participants were able to complete tasks and had positive feedback regarding the difficulty, visibility, efficiency, and integration of our design (Fig. 5, 7, 9, 11). However, one of the pitfalls of our design was that we did not consider unexpected task flows that users took to complete a task during usability testing and thus did not design to account for that sequence of events. This was evidenced by the significantly larger number of clicks it took for some participants to complete a task (Fig. 2-4). Some participants noted that some features of the design were unintuitive or difficult when completing core tasks – notably the 'search for recipes' task and 'mark food items as complete' task.

After validating our overall approach using usability testing, we identified some issues that participants experienced, particularly the less than ideal visibility of certain features such as changing the expiry date, confusion of what the check mark buttons on pantry page leads to, and the 'search for recipes' feature (Fig. 12). Overall, our design was shown to function well with the overwhelming majority (all but one) of participants being able to complete all tasks in the usability test. This meant our features were working as intended but we did note that some participants expressed concern on the intuitiveness and visual clutter of the 'search for recipes' feature, stating difficulties in use cases where they had to search for a specific ingredient or when there were a lot of items in your pantry (Fig. 6). Similarly, some participants reported that they didn't know if the check mark beside each food item on the pantry page was meant to mark the item as complete or was there to select the item for recipe search. As such, we recommend that our prototype design be iterated further in the following ways based on the feedback provided:

1. Differentiate 'mark as complete' button by changing the check icon and add a 'select' button
2. Color the filters on 'find recipes' page and remove the 'X' icon beside the label to reduce clutter
3. Highlight 'edit' button on 'find recipes' as main method to select pantry items to improve clarity

These changes to our design should ensure that the features we offer are more functional, intuitive, and easier to use. Additionally, we recommend minor adjustments to be made to the overall design to improve its learnability whereby it no longer requires guidance of another to user, as was seen with one participant who had to rely on the researcher to explain it to them during the usability test (Fig. 8). Following our recommended design changes, we suggest conducting another formative study to validate that the data supports resolution of the feedback/issues identified, and that users are still able to complete core tasks.

Critique of Process

Our chosen methods provided valuable data; the observation provided good insight into users' thought processes, and the questionnaire allowed users to give comprehensive feedback. However, the prototype could have been improved. When presenting it to users, we separated the tasks into chunks, like chapters in a game, to guide users through each task. However, this approach made the interaction of our app less natural, since users were restricted to performing one set of actions, and there were no possibilities for exploration. Additionally, some issues were caused by designer bias, as we falsely assumed that users would perform actions the same way we, as designers, would. We piloted a CPSC 344 student but because they had sufficient background knowledge, they did not experience the issues that someone with less design expertise may experience. Another pitfall was that although we designed our process to measure efficiency, our test protocol led us to inadvertently evaluate learnability rather than efficiency. There were contradictions between objective and subjective data corresponding to our analysis (Fig. 5,6); many participants had a high number of clicks when performing tasks, indicating our design was inefficient, yet felt that the design was efficient. This may be due to unfamiliarity with our prototype, as participants were not shown how to use the design beforehand, resulting in the novelty effect. This was an intentional decision; we felt that if we had shown the participants how to use our prototype prior, they would perform actions the same way and skew the data. In hindsight, we should have instead allowed users to explore the prototype before the usability test to avoid both experimenter bias and novelty effect.

Appendix A: https://www.figma.com/design/eK8l3Sku1jEgBMSwGVlH7b/CPSC-344-T1H?node-id=574-8408&t=31bsrN8eJ5B4UmTb-1

Appendix B: Design Outcomes Written Final Report